

Aerothermochemical analysis of the Orion capsule in rarefied transitional flow regime during Mars entry

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Abstract

It is well known that NASA developed the Orion project, besides for a rescue shuttle for the International Space Station, also for the transportation of instruments and astronauts to Moon, Mars and beyond. The forthcoming use of Orion for Mars landing stimulated Zuppardi to compute global aerodynamic coefficients in rarefied, transitional flow along a high altitude entry path. A Direct Simulation Monte Carlo (DSMC) code computed the coefficients at the altitudes of 80, 90 and 100 km, in the range of angles of attack 0–180 deg. The aim of the present paper, that can be considered as the logic follow on of the previous paper by Zuppardi, is to evaluate, in high altitude entry path, the influence of: 1) chemical reactions on the flow field and on some local quantities (temperature, pressure and heat flux) along the Orion surface, 2) typology of entry trajectory (direct and orbit) on both local quantities and drag coefficient, 3) **heat flux associated to the surface recombination reactions**. Computations were carried out in axisymmetric flow and in the altitudes interval 60–100 km by means of a home made code for the computation of the entry trajectories and of the powerful and advanced two-dimensional/axisymmetric DSMC code (DS2V) for the solution of the flow field around Orion.

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1. Introduction

Orion is a manned, partially reusable capsule for future space exploration. It was conceived as a rescue shuttle for the International Space Station (ISS). According to the NASA [Artemis program \(2020\)](#), Orion will be used to transport instruments and astronauts in future manned lunar and Martian missions. NASA originally started this project to replace the retired Space Shuttle as a link to/from the ISS.

Orion is similar in shape but larger than Apollo Command Module (CM). Indeed, diameter and mass of Orion

are 5.03 m and 8500 kg, while diameter and mass of Apollo CM are 3.90 m and 5557 kg. Orion can transport cargo and four to six astronauts for as many as 21 days undocked and up to six months docked.

NASA is currently still developing the Orion project. The first, unmanned flight test took place on Dec. 5, 2014. Orion was launched from Cape Canaveral, reached an altitude of 5700 km, made two terrestrial orbits and finally splashed down into Pacific Ocean off the coast of California. Aerodynamics of Orion in Earth atmosphere entry from ISS was already studied by [Moss et al. \(2006\)](#) in all hypersonic rarefied regimes i.e. from free molecular flow to continuum. Moss carried out the calculations in rarefied flow by two Direct Simulation Monte Carlo (DSMC) codes and in continuum by a Computational Fluid Dynamic (CFD) code. [Zuppardi \(2021\)](#) computed

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Nomenclature

C_D, C_h, C_p	drag, heat flux, pressure coefficients	R_N	Orion nose radius, m
CFD	Computational Fluid Dynamics	S	Orion base area, m^2
CG	Center of Gravity	s	curvilinear abscissa, m
CM	Command Module	T	temperature, K
c_p	specific heat at constant pressure, J/kg/K	TPS	Thermal Protection System
D	Orion base area diameter, m or drag force, N	t	time, s
DSMC	Direct Simulation Monte Carlo	V	velocity, m/s
FC	Fully Catalytic surface	X, Y	dimensions of computational domain, m
g	gravity acceleration, m/s^2	X_S	molar fraction of the chemical species S
H	specific total enthalpy, J/kg	x, y	Cartesian coordinates, m
h	altitude, km	γ	flight path angle, deg
IN	inert gas	ε	emissivity
IS	insulated surface	λ	mean free path, m
ISS	International Space Station	ρ	density, kg/m^3
Kn	Knudsen number		
L	Orion length, m	<i>Subscripts</i>	
Ma	Mach number	∞	free stream
m	Orion mass, kg	c	convective
mcs	mean collision separation, m	f	fluid dynamic
N	number density, $1/m^3$	r	recombination
NC	Non Catalytic surface	s	simulation
p	pressure, Pa	w	surface
$\dot{q}$	heat flux, W/m^2	2	downstream of a normal shock wave
R	trajectory curvature radius, m		
Re	Reynolds number		

the global aerodynamic coefficients along an entry path at the altitudes of 80, 90 and 100 km, in the range of angle of attack 0–180 deg and fulfilled a sensitivity analysis of longitudinal moment and equilibrium angle of attack in terms of position of the center of gravity.

The aim of the present paper is to evaluate, at the beginning of the entry path, the influence of:

1. chemical reactions on the flow field and, more specifically, on temperature around the capsule as well as the surface catalytic effects on heat flux,
2. typology of entry trajectory (direct and orbit) on local quantities (such as temperature, pressure and heat flux), on the drag and on the related coefficients.
3. heat flux associated to the surface recombination reactions (or surface catalyticity).

Such local quantities were also analyzed at the stagnation point which, as well known, is the most stressed point of a spacecraft during entry/reentry.

Such a kind of analysis in hypersonic, rarefied re-entry flows is not recent and is still in progress. See, for example, the study of thermo-chemical, non-equilibrium flow around: i) American Stardust capsule (Boyd et al., 2007) to collect samples of material from comet Wild-2. This capsule recorded the fastest man-

made spacecraft re-entry speed of 12.8 km/s, ii) Chinese Shenzhou lunar exploration capsule (Feng et al., 2018), launched in 2014 and returned to Earth with a speed of 11.2 km/s, iii) Tiangong-1 Chinese target spacecraft (Li et al., 2019).

The present computations were carried out in the altitude interval 60–100 km and in axisymmetric flow or at zero angle of attack. The choice of zero angle of attack is reasonable because Orion, as shown later, is a revolution body. Zero angle of attack maximizes the surface exposed to the flow and should maximize the aerodynamic drag, thus the capsule deceleration. It has to be pointed out that the trim angle of Orion is different from zero. However, the choice of zero angle of attack is proper for the purpose of this paper because this angle of attack produces the most severe conditions in terms of the above mentioned local quantities.

A home made code (Zuppari and Savino, 2015) computed the two Mars entry trajectories, direct and orbit, and a DSMC code DS2V (Ver. 4.5, 64 bits, Bird, 2008) computed the axisymmetric flow field around Orion. Computations were performed considering the two trajectories and four conditions: 1) inert gas, 2) chemically active gas and non catalytic surface, 3) chemically active gas and fully catalytic surface, 4) chemically active gas and insulated surface.

2. Direct simulation Monte Carlo method and DS2V code

The Direct Simulation Monte Carlo (DSMC) method (Bird, 1998, 2013, Shen, 2005) is a stochastic method, solving flow fields in transitional regime i.e. from continuum low density to free molecule regimes. As well known, the DSMC method is an indispensable tool for solving rarefied flow fields. In such conditions, using the Navier-Stokes equations is not correct because of the failure of the Chapman-Enskog theory for the computation of transport coefficients. On the other hand, the solution of the Boltzmann equation is very difficult. Furthermore, the Boltzmann equation can not consider chemical reactions; these are very important in highly energized flows like those met along an entry path.

Fundamentals of the method are now well known and widely accepted by the scientific community so they will be not described here. On the opposite, some computing procedures, making a DSMC code more efficient and accurate than a “basic” code, will be here described. A DSMC code, implementing these procedures, like the two-dimensional/axisymmetric DS2V code, is labeled in literature as DSMC07 (Bird, 2006; Bird et al., 2009; Gallis et al., 2009). A basic code is labeled as DSMC94. More specifically, DS2V:

- (1) divides the computation volume into two grids of cells (collision and sampling) with the possibility of adapting the cells. DS2V automatically generates the number of cells and the structure of both collision and sampling grids on the basis of the input number of megabytes and of the free stream number density. In the present computations, the number of megabytes was about 1250. The user can change, during a computation, the number of cells and therefore the structure of both grids by inputting the numbers of molecules for the adaptation of collision and sampling cells; the lower the input number of molecules per cell, the higher the number of cells after the adaptation process. In the present computations, the adaptation procedure used 8 and about 200 molecules for collision and sampling cells, respectively.
- (2) provides an optimal global time step. This is computed as a fraction (say 0.2) of the current time step in each collision cell, averaged over the computational domain. This time step is the smallest between the collision time and the transit time or the time necessary to a molecule for transiting the collision cell at its own velocity. In a DSMC94 code, the global time step is an input datum, kept constant during the evolution of the system, therefore irrespective of the actual, evolving fluid dynamic conditions.
- (3) avoids sequential collisions between the same molecules. This, in fact, would be physically impossible because the molecules, after a collision, move away in opposite directions.

- (4) allows the user to evaluate the quality of a computation by means of the evaluation of the adequacy of the numbers of simulated molecules and of collision cells. The ratio between the molecule mean collision separation (mcs) and the mean free path (λ), in each collision cell, is the parameter that makes this evaluation possible; mcs/λ has to be less than unity everywhere in the computational domain for a good quality of a computation. Bird (2006) suggests that mcs/λ should be less than 0.2 for an optimal quality of a DSMC computation. DS2V outputs the value of mcs/λ , both averaged over the computational domain and the maximum value.
- (5) provides indication about the stabilization of a computation. Stabilization is achieved when the profile of the number of simulated molecules, as a function of the simulated time, becomes jagged and included within a band that represents the standard deviation of the number of simulated molecules.

3. Orion geometry

Orion is a revolution body whose geometry (Moss et al., 2006) in the meridian plane (x, y) is shown in Fig. 1. Maximum diameter is $D = 5.03$ m; therefore, the base area or the area of the maximum cross section is $S = 19.86$ m². S is used as reference area for scaling the (axial) force to the drag coefficient. The curvature radius of the rounded shoulder, i.e. the curve between points C and D, is 0.25 m. The center of curvature is located at $x = 0.71$ m, $y = 2.26$ m. The curvature radius of the heat shield (which is also the radius of the nose R_N) is 6.04 m, the center is located on the x axis. The center of gravity (CG) is located at $x_{CG} = 1.33$ m, $y_{CG} = 0.24$ m (Moss et al., 2006), i.e. not on the x -axis therefore the Orion capsule has a non-zero trim angle of attack. For example, Andersen and Whitmore (2007) computed, by means of a CFD code, a trim angle of about -20 deg, considering no flap and in continuum Earth re-entry.

The computational domain was a rectangle in the meridian plane: $X = 6.0$ m, $Y = 3.5$ m. The capsule surface was approximated in 480 intervals of which 350 were for the Thermal Protection System (TPS), i.e. between points C and E.

4. Mars atmosphere and entry trajectories

The present computations considered the Mars atmosphere made up of 7 species and its composition constant with altitude. Table 1 reports the mass and the molar fractions of the chemical species.

Due to dissociation along an entry path, atomic oxygen and atomic nitrogen are also present thus, in these simulations, the working gas is a mixture of 9 chemical species. The chemical model, proposed by Bird in the version 3.3 of DS2V code (Bird, 2005), has been implemented also in

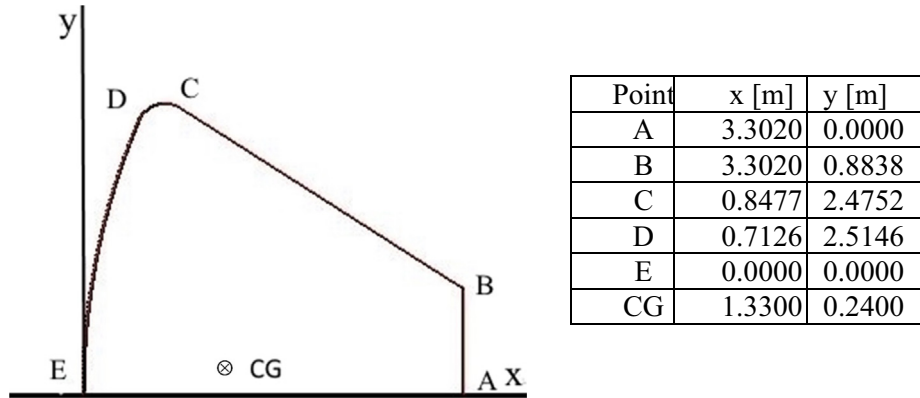


Fig. 1. Geometry of Orion in the meridian plane (x, y).

the present version (4.5). This model is made of 54 reactions: 40 dissociations, 7 forward (or endothermic) exchanges, 7 reverse (or exothermic) exchanges.

The Mars Global Reference Atmospheric Model 2001 (GRAM-2001, Justus and Johnson, 2001) provided the atmosphere parameters. The parameters (temperature, density therefore number density) used in the present computations are those reported by Justus and Johnson in the altitude interval $-1.365 \div 194.654$ km with a step of about 5 km. The parameters at the intermediate altitudes in each step were computed by linear interpolation.

The entry trajectories of Orion were computed by the integration of the equations of the capsule dynamics (Zuppardi and Savino, 2015). As said before, the capsule was considered at zero angle of attack along the whole entry path i.e. with no lift and in free entry or with no thrusters and no parachutes:

$$\frac{dV}{dh} = \frac{1}{2} \frac{\rho V}{\sin \gamma} \frac{SC_D}{m} - \frac{g}{V} \quad (1.a)$$

$$\frac{d\gamma}{dh} = \frac{1}{R \sin \gamma} - \frac{g}{V^2} - \frac{1}{tg \gamma} \quad (1.b)$$

being: V velocity, ρ density, γ flight path angle, g gravity acceleration, R curvature radius of the trajectory, m capsule mass, C_D drag coefficient. The ratio m/SC_D is the ballistic parameter.

Integration of Eqs. 1(a), 1(b) makes possible the computation of entry trajectory (or velocity) as function of altitude. The code provides both input data to DS2V and free stream Mach and Reynolds numbers as well as Rey-

nolds number downstream a normal shock wave. Equations were integrated numerically by a forward scheme with a first order approximation (Euler method). Integration started from $h = 100$ km and the step was 0.1 km. The initial data of the integration procedure are reported in Table 2 for direct and orbit entry trajectories. The initial values of speed and angle of entry are those of Pathfinder for the calculation of the direct reentry trajectory and those of Viking 1/2 for the calculation of the orbit reentry trajectory (Braun and Manning, 2006). In order to consider the proper values of the drag coefficient at each altitude, the solution of Eqs. 1(a), 1(b) has to rely on the iterative procedure already used by Zuppardi (2019) in computing the Pathfinder entry trajectory in Mars atmosphere. The first step is computing a preliminary entry trajectory using $C_D = 1.0$ at each altitude. This step produces input data to a CFD and/or DSMC code for the computation of the updated values of the drag coefficient at each altitude. This step is iterated until the entry trajectory no longer changes significantly. Zuppardi verified, in the study of the entry trajectory of Pathfinder, that the procedure can stop at the first step; the same has been made in the present computations.

Fig. 2(a), 2(b) and 2(c) show the profiles of free stream velocity, Mach and Reynolds numbers as functions of altitude. Table 3 reports the free stream input data to DS2V: temperature (T_∞), number density (N_∞). Free stream density (ρ_∞), pressure (p_∞) and Knudsen number ($Kn_{\infty D}$) are also reported. Tables 4(a) and 4(b) report for both direct and orbit entry: free stream velocity (V_∞), Mach number (Ma_∞), Reynolds number ($Re_{\infty D}$) and Reynolds number downstream a normal shock wave (Re_{2D}).

Table 1
Chemical composition of Mars atmosphere.

Chemical species	Mass fraction	Molar fraction
O ₂	0.0013	0.00176
N ₂	0.0270	0.04173
NO	0.0001	0.00014
CO	0.0007	0.00108
CO ₂	0.9500	0.93399
C	0.0049	0.00396
Ar	0.0160	0.01734

Table 2
Orion operative parameters at $h = 100$ km.

Entry trajectory	Direct	Orbit
Entry velocity, m/s	7260	4700
Entry Path Angle, deg	14.06	17.00
Entry mass, kg	8500	8500
Entry ballistic parameter, kg/m ²	214	214
Entry angle of attack, deg	0	0

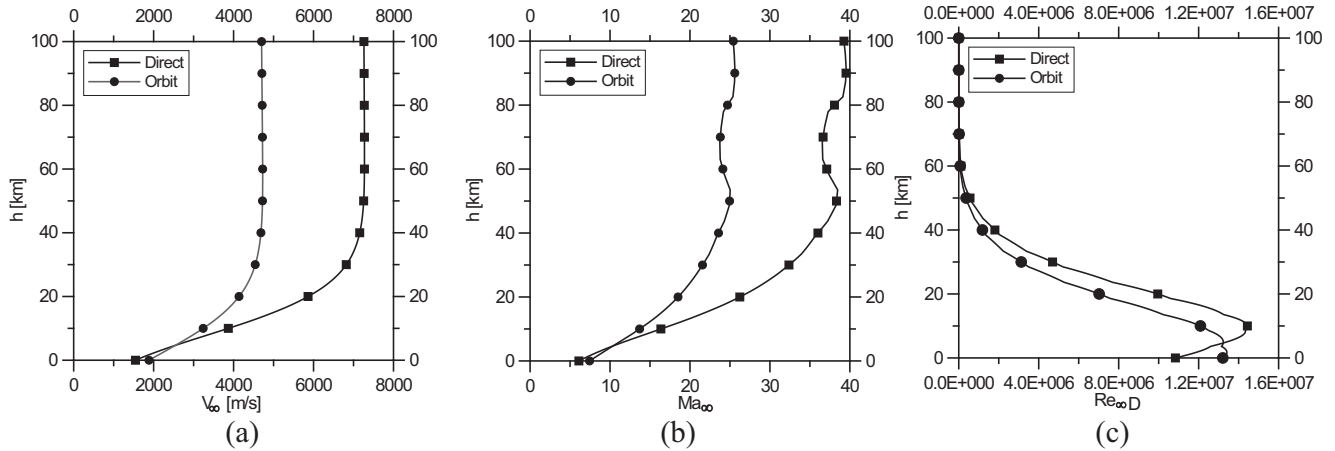


Fig. 2. Profiles of Orion free stream velocity (a), Mach number (b), Reynolds number (c).

Table 3

Input data to DS2V, free stream density, pressure and Knudsen number.

h [km]	T_∞ [K]	N_∞ [1/m ³]	ρ_∞ [kg/m ³]	p_∞ [Pa]	$Kn_{\infty D}$
100	127	8.05×10^{17}	5.88×10^{-8}	1.48×10^{-3}	1.69×10^{-1}
90	125	3.66×10^{18}	2.67×10^{-7}	6.70×10^{-3}	3.69×10^{-2}
80	135	1.94×10^{19}	1.42×10^{-6}	3.74×10^{-2}	7.20×10^{-3}
70	146	8.09×10^{19}	5.92×10^{-6}	1.64×10^{-1}	1.78×10^{-3}
60	143	3.20×10^{20}	2.34×10^{-5}	6.18×10^{-1}	4.46×10^{-4}

Table 4a

Free stream velocity and some aerodynamic parameters for direct entry.

h [km]	V_∞ [m/s]	Ma_∞	$Re_{\infty D}$	Re_{2D}
100	7260	39.3	3.17×10^2	3.77×10^0
90	7265	39.5	1.46×10^3	1.71×10^1
80	7269	38.1	7.20×10^3	9.07×10^1
70	7272	36.7	2.80×10^4	3.79×10^2
60	7270	37.1	1.13×10^5	1.50×10^3

Direct entry velocity is about 50% higher than orbit entry velocity. The same occurs for free stream Mach and Reynolds numbers. For both entry trajectories, velocity and Mach number do not change meaningfully in the altitude interval. On the opposite, Reynolds number reduces by three orders of magnitude and Knudsen number increases by four orders of magnitude. According to Vallerani (1973), the transitional regime for a blunt body is defined in terms of Reynolds number downstream a normal shock wave, by $10^{-1} < Re_{2D} < 10^4$. According to Moss

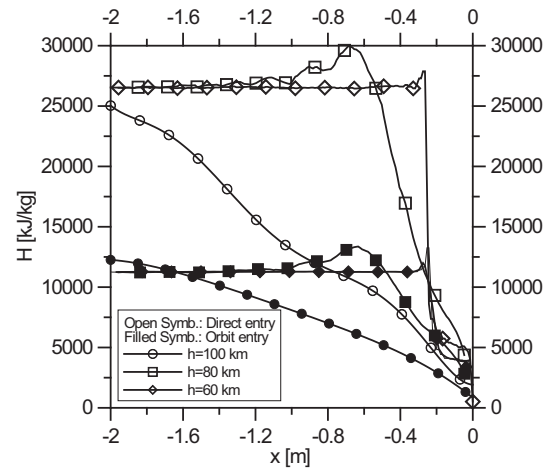


Fig. 3. Profiles of total enthalpy along the Orion stagnation line.

(1995), it is defined in terms of global Knudsen number by $10^{-3} < Kn_{\infty D} < 50$. Table 3 shows that Re_{2D} and $Kn_{\infty D}$ agree in stating that the flow field is in transitional regime.

Table 4b

Free stream velocity and some aerodynamic parameters for orbit entry.

h [km]	V_∞ [m/s]	Ma_∞	$Re_{\infty D}$	Re_{2D}
100	4700	25.4	2.00×10^2	2.32×10^0
90	4707	25.6	9.20×10^2	1.05×10^1
80	4715	24.7	4.55×10^3	5.57×10^1
70	4722	23.8	1.77×10^4	2.32×10^2
60	4727	24.1	7.17×10^4	9.17×10^2

5. Test conditions

For all DSMC computations, the gas-surface interaction model was **diffusive, fully accommodated** i.e. in equilibrium with surface temperature (T_w). In order to evaluate the chemical effects, computations were carried out considering only the direct entry trajectory because, as shown in Fig. 3,

the flow field is much more energized than that of the orbit entry. Flow energy is quantified by the total enthalpy H ($H = V^2/2 + c_p T$, J/kg, $c_p = 667.6$ J/kg/K). H , along a direct entry trajectory, is more than twice than that along an orbit entry trajectory.

Computations were carried out to evaluate the contribution of the surface recombination reactions on heat flux (surface catalyticity). For both non catalytic (NC) and fully catalytic (FC) surface, the surface temperature ($T_w = 300$ K) was constant along the whole capsule. Even though this surface temperature is not realistic for an entry trajectory, it is a conventional value to evaluate thermal problems along an entry trajectory. The five surface recombination reactions, reported in Table 5 with the related exothermic energies (taken from Bird, 2005), were implemented. DS2V carries out the surface recombination reactions on the base of a pre-stated probability; the conditions of FC and NC surface are simulated by fixing this probability to unity and to zero, respectively. When a molecule impacts onto a surface, it remains on the surface. If a second molecule, that impacts onto the same elemental area, can recombine with the first one, they do and the recombined molecule returns to the flow, otherwise each molecule leaves the surface separately. If a recombination reaction happens, all energy from the reaction is transferred to the surface thus heat flux increases. Therefore, heat flux includes convective heat flux (kinetic and internal energies) and heat flux from surface catalytic reactions, if any: $\dot{q} = \dot{q}_c + \dot{q}_r$.

To evaluate chemical and thermal effects, four sets of simulations were carried out at each altitude: 1) chemically inert gas (IN), 2) chemically active gas and non catalytic surface (NC), 3) chemically active gas and fully catalytic surface (FC), 4) chemically active gas and insulated surface (IS). The second and third test cases were carried out to evaluate the influence of the surface catalyticity on heat flux. The fourth test case was carried out for a preliminary evaluation of the profile of surface temperature. For this case, DS2V considers the surface with zero thermal conductivity and takes into account, in thermal balance, also thermal radiation from the surface. The surface emissivity (ϵ) is required in input for each elemental area. In the present computations, the realistic value of $\epsilon = 0.8$, constant along the capsule, was used. According to what said before, only the NC and IS surface were considered for the orbit entry computations.

Table 5
Surface recombination reactions and exothermic energy.

Surface reactions	Molecule [J]	Specific [J/kg]
#1: $O + O \rightarrow O_2$	8.197×10^{-19}	1.543×10^7
#2: $N + N \rightarrow N_2$	1.561×10^{-18}	3.357×10^7
#3: $N + O \rightarrow NO$	1.042×10^{-18}	2.089×10^7
#4: $CO + O \rightarrow CO_2$	7.248×10^{-19}	9.915×10^6
#5: $C + O \rightarrow CO$	1.781×10^{-18}	3.830×10^7

6. Quality of results

Tables 6a and 6b verify the quality of the DS2V results for direct and orbit trajectories, respectively. The value of mcs/λ , averaged over the flow field, is less than unity at each altitude and the Bird's criterion for an optimal quality of a DSMC computation ($mcs/\lambda \leq 0.2$, Bird, 2006) is fully satisfied at altitudes higher than 70 km.

A steady state condition is also achieved. In fact, as well known, a rule of thumb suggests considering an unsteady, fluid dynamic computation stabilized when the ratio t_s/t_f is reasonably greater than unity, being t_s the simulation time and t_f the fluid dynamic time. The latter is computed as the time necessary for the fluid to cross the length (L) of the body under study at the free stream velocity: $t_f = L/V_\infty$ (for the present case, $L = 3.30$ m). Increasing t_s is beneficial because the longer t_s the larger the number of samples of the molecular quantities for the computation of the macroscopic fluid dynamic quantities. Furthermore, increasing the number of samples is equivalent to a calculation with a larger number of simulated molecules, making possible in principle, that the fluctuations are the same as those in the real gas.

7. Analysis of the results

7.1. Effects of chemistry

The comparison of Fig. 4(a) and 4(b) provides a qualitative evaluation of the effects of chemistry (or of dissociation) in terms of temperature in the flow field. This evaluation has been made at altitude of $h = 60$ km where the chemical effects are more relevant compared with those at higher altitudes. As expected, due to the endothermic nature of the dissociation reactions, temperature downstream a shock wave is higher when gas is considered chemically inert.

Fig. 5(a) shows the profiles of temperature along the Orion stagnation line computed at the three altitudes with and without chemistry. Fig. 5(b) shows the effects of chemistry by the relative increase of temperature: $\Delta T/T = (T_{IN} - T)/T$, being T_{IN} and T local temperatures calculated considering gas inert and chemically reactive. As expected, the maximum value of $\Delta T/T$ is obtained at $h = 60$ km and is located in the shock wave.

The effect of chemical reactions or of dissociation can also be evaluated by the profiles of the percentage molar fraction of the greatest component of Martian atmosphere (CO_2) and of the products of its dissociation. Fig. 6(a) and 6(b) show the profiles of $X_{CO_2}\%$ and, for example, of $X_C\%$ along the Orion stagnation line at the three altitudes. The profiles of $\Delta T/T$, $X_{CO_2}\%$ and $X_C\%$ reproduce the profile of a shock wave. In fact, like the shock wave thickness, these profiles are narrow at lower altitude ($h = 60$ km) and gradually widen with increasing altitude.

Fig. 3 already verified the beneficial effect of dissociation. In fact, since dissociation reactions are endothermic,

Table 6a

Quality of computations for direct entry trajectory.

h [km]	IN		NC		FC		IS	
	mcs/ λ	t_s/t_f	mcs/ λ	t_s/t_f	mcs/ λ	t_s/t_f	mcs/ λ	t_s/t_f
100	0.001	9.53	0.001	8.28	0.001	8.78	0.001	9.15
90	0.003	8.90	0.003	7.63	0.003	8.33	0.003	8.53
80	0.013	7.17	0.017	5.43	0.014	5.44	0.015	5.81
70	0.045	5.85	0.069	4.38	0.063	3.47	0.067	5.74
60	0.182	3.32	0.285	4.54	0.256	2.76	0.257	3.04

Table 6b

Quality of computations for orbit entry trajectory.

h [km]	NC		IS	
	mcs/ λ	t_s/t_f	mcs/ λ	t_s/t_f
100	0.001	7.39	0.001	8.95
90	0.003	8.39	0.003	9.24
80	0.016	6.10	0.014	6.79
70	0.071	3.61	0.064	3.93
60	0.266	2.45	0.263	2.31

H decreases at each altitude by one order of magnitude. The amount of energy per unit time and surface (or heat flux $\dot{q}_r$ [W/m²]), supplied by the five surface recombination reactions as functions of altitude is shown in Fig. 7. Reactions #1 ($O + O \rightarrow O_2$) and #4 ($CO + O \rightarrow CO_2$) are the main contributors to the surface catalytic heat flux at each altitude. The resultant heat flux provided to the whole surface by all reactions decreases by four orders of magnitude from $h = 60$ to $h = 100$ km. Table 7 reports the stagnation point heat flux for FC and NC surface at each altitude. Heat fluxes increase by two orders of magnitude from $h = 100$ km to $h = 60$ km. The percentage variation between the two sets of heat flux can be considered as a measure of catalyticity. As expected, due to a decrease of the rate of recombination reactions with rarefaction, the catalytic effect decreases with altitude.

7.2. Effects of the entry trajectory

Fig. 8(a), 9(a), and 10(a) show the profiles of temperature, pressure and heat flux along the Orion TPS, at the altitudes of $h = 60, 80$ and 100 km for the two entry trajectories. The corresponding Fig. 8(b), 9(b) and 10(b) show the profiles of the same quantities at the stagnation point as functions of altitude in the interval 60–100 km. Temperature increases from $h = 100$ km to $h = 60$ km by one order of magnitude, pressure and heat flux by three orders of magnitude. Heat flux at the stagnation point computed considering NC, FC surface and direct entry, already reported in Table 7, are shown also in Fig. 10(b). For the sake of completeness, the convective heat flux by DS2V has been compared with the convective heat flux from the engineering formula by West and Brandis (2020):

$$\dot{q}_c(0) = 7.207 \times 10^4 \rho^{0.47} R_N^{-0.54} V_\infty^{3.5} \quad (2)$$

where V_∞ is the free stream velocity in km/s, ρ is the free stream density in kg/m³ and R_N is the radius of the nose in m. West and Brandis obtained Eq.2 by correlating 390 CFD computations in a wide interval of free stream density (from 10^{-5} to 10^{-3} kg/m³), nose radius (from 1 to 20 m) and free stream velocity (from 2 to 8 km/s). This formula computes the convective heat flux at the stagnation point of an axisymmetric body in Mars entry. The comparison

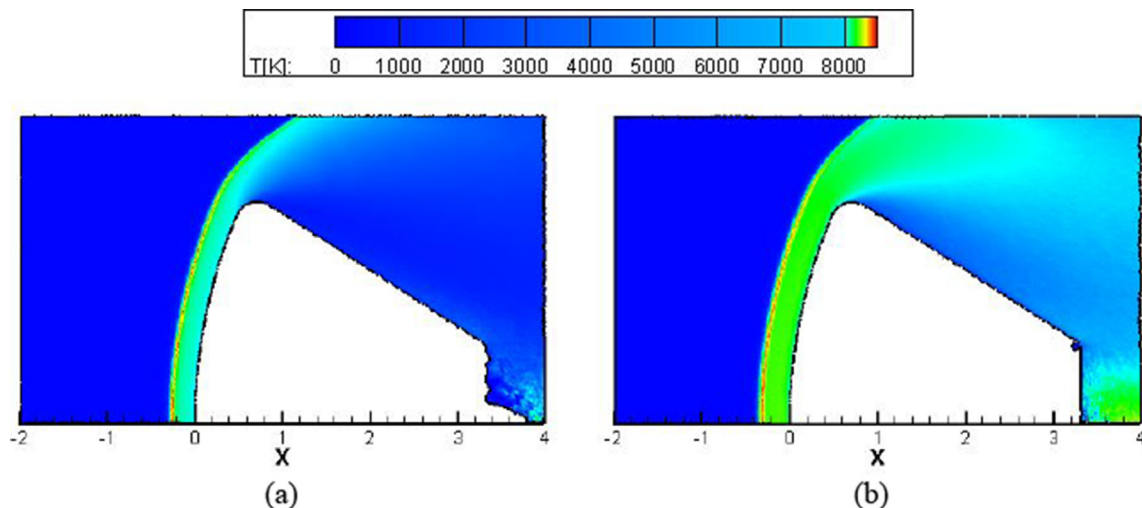


Fig. 4. Two-dimensional maps of temperature in the flow field around Orion with (a) and without (b) chemical reactions: $h = 60$ km, direct entry, NC surface.

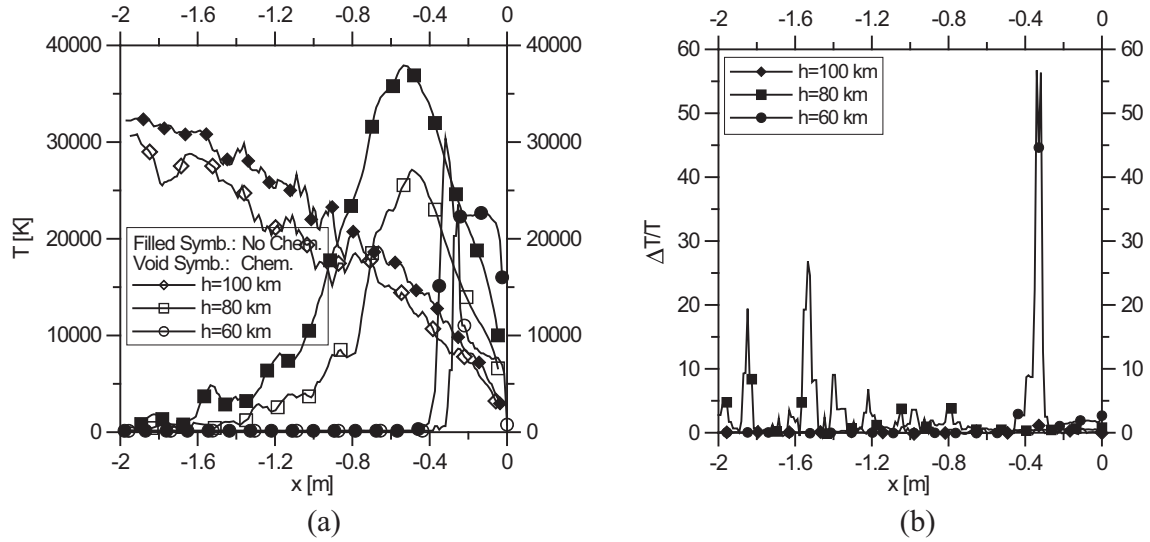


Fig. 5. Profiles of temperature along the Orion stagnation line with and without chemistry (a) and of relative increment along the stagnation line (b): NC surface, direct entry.

of the convective heat flux is excellent for direct entry and reasonably good for orbit entry.

As expected, higher velocity by direct entry involves higher values of local quantities (temperature, pressure and heat flux) and of Mach and Reynolds numbers. However, as shown in Fig. 11(a), 11(b) and 11(c), the values of the corresponding coefficients, as well as the drag coefficient, do not present meaningful variations. Drag (C_D), heat flux (C_h) and pressure (C_p) coefficients are, by definition:

$$C_D = D / (0.5 \rho_\infty V_\infty^2 S) \quad (3.a)$$

$$C_h = \dot{q} / (0.5 \rho_\infty V_\infty^3) \quad (3.b)$$

$$C_p = (p - p_\infty) / (0.5 \rho_\infty V_\infty^2) \quad (3.c)$$

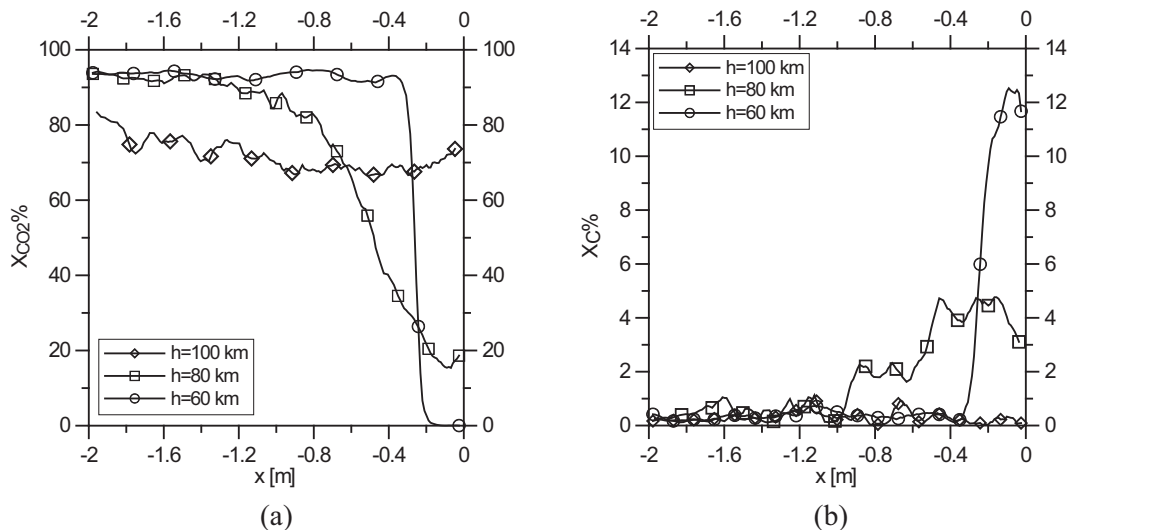


Fig. 6. Profiles of CO_2 (a) and C (b) molar fractions along the Orion stagnation line at three altitudes: direct entry.

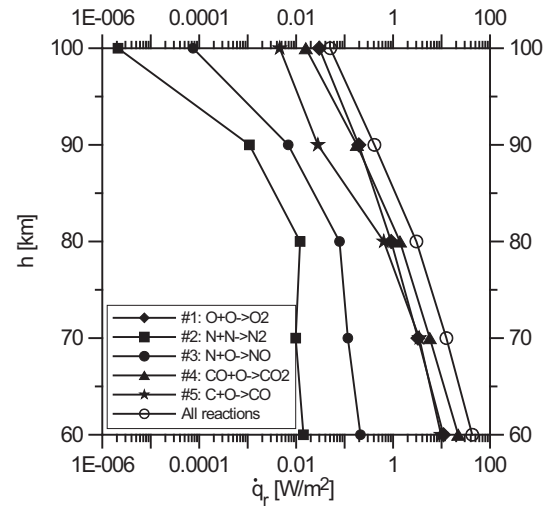


Fig. 7. Profiles of energy for unit time and unit surface from each recombination reaction and from all reactions.

Table 7

Heat flux at the Orion stagnation point for FC and NC surface and percentage difference: direct entry.

h [km]	$\dot{q}(0)_{FC}$ [kW/m ²]	$\dot{q}(0)_{NC}$ [kW/m ²]	$(\dot{q}(0)_{FC} - \dot{q}(0)_{NC})/\dot{q}(0)_{NC} \%$
100	9.0	8.5	5.9
90	30.0	23.7	26.2
80	71.4	37.8	88.9
70	217.1	72.2	200.7
60	706.2	193.1	265.8

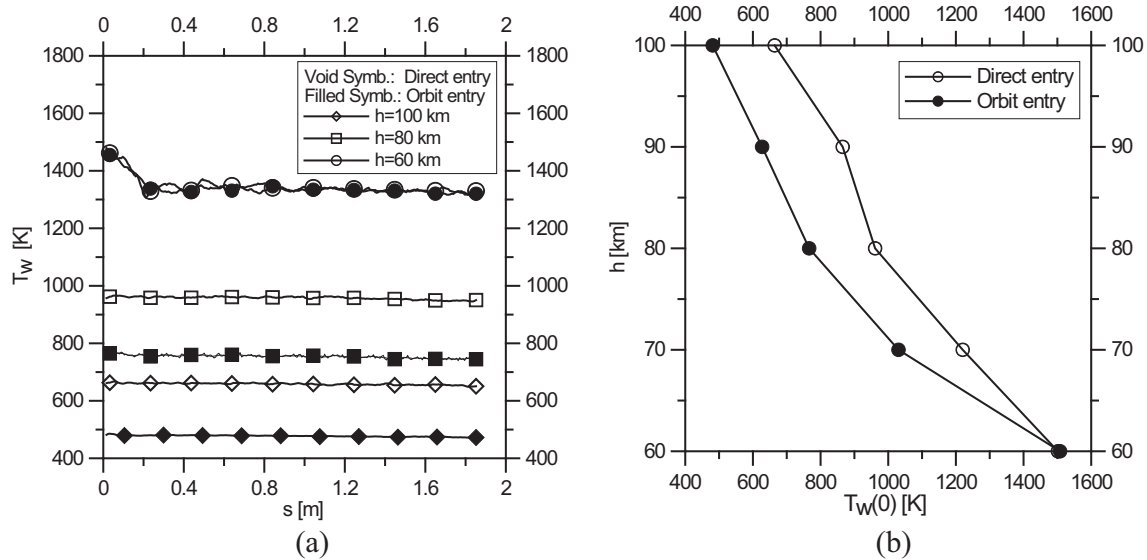


Fig. 8. Profiles of temperature along the Orion heat shield surface (a) and at the stagnation point as function of altitude (b).

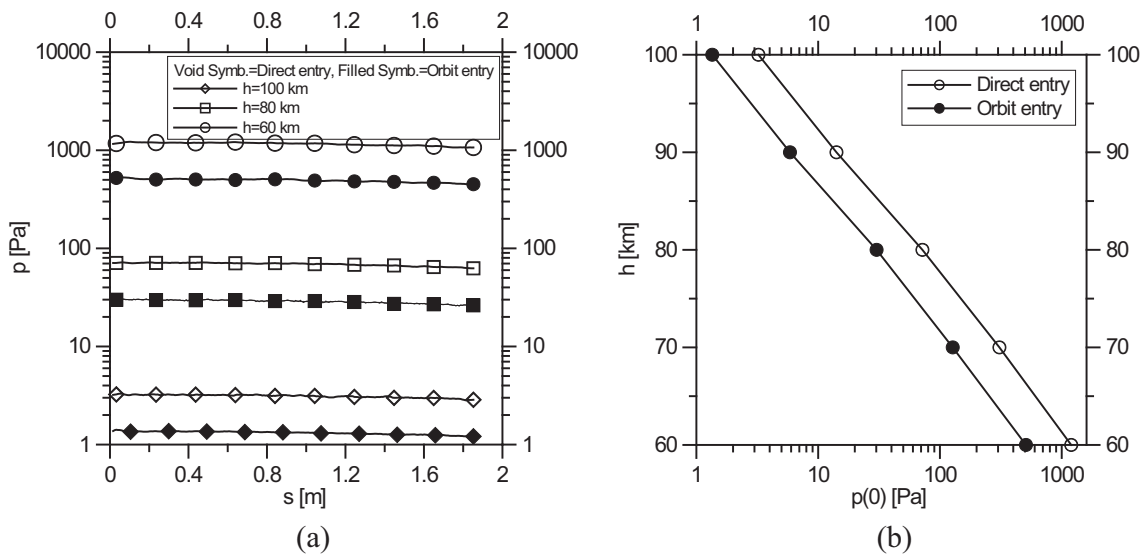


Fig. 9. Profiles of pressure along the Orion heat shield surface (a) and at the stagnation point as function of altitude (b).

Since, as reported in Tables 4(a) and 4(b), Mach number is higher than 20, the low variability of the coefficients between the two entry trajectories is certainly due to the

Mach number independence principle (Anderson, 1989). Table 8 quantifies what said in terms of drag force and of drag coefficient as functions of altitude.

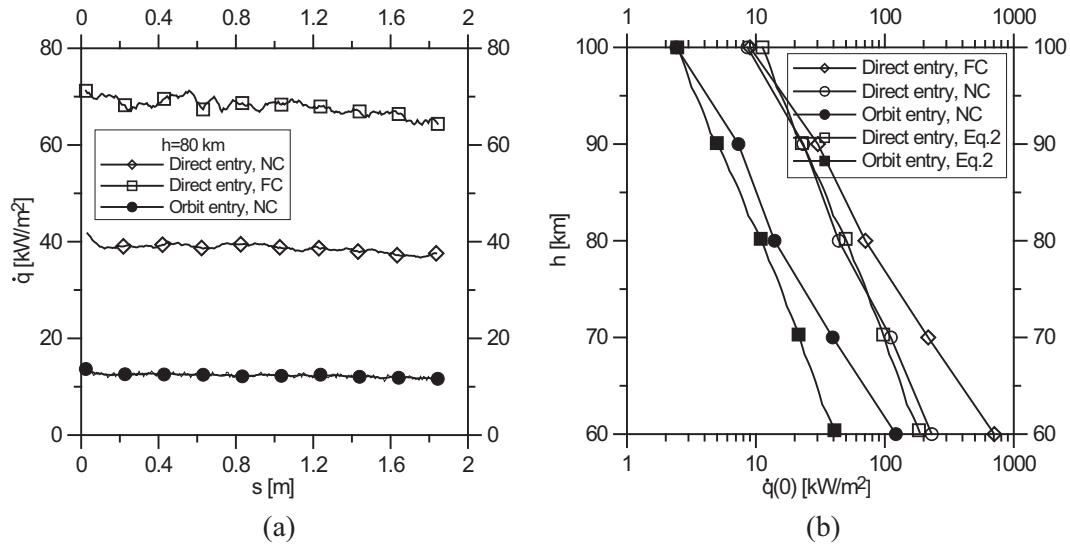


Fig. 10. Profiles of heat flux along the Orion heat shield surface at $h = 80$ km (a) and at the stagnation point as function of altitude (b).

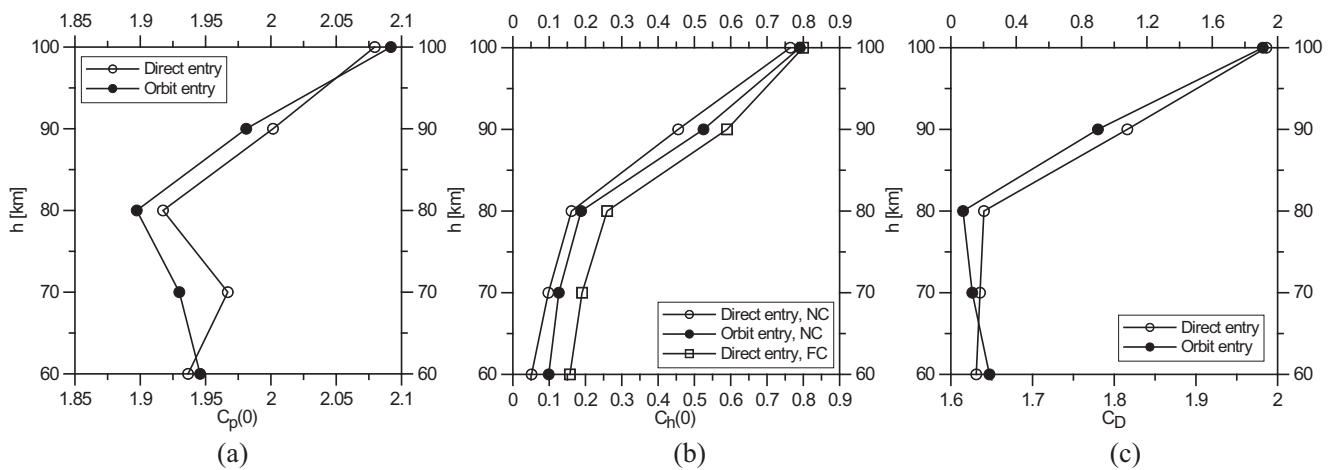


Fig. 11. Profiles of the pressure (a), heat flux (b) at the Orion stagnation point and Orion drag (c) coefficients as functions of altitude.

Table 8

Orion drag, drag coefficient and related percentage differences as functions of altitude.

h [km]	D [N] Direct entry	D [N] Orbit entry	$\Delta D/D\%$	C_D Direct entry	C_D Orbit entry	$\Delta C_D/C_D\%$
100	61.2	25.6	139.1	1.99	1.98	0.22
90	254.4	104.7	143.0	1.82	1.78	2.02
80	1219	504.8	141.5	1.64	1.62	1.59
70	5083	2130	138.6	1.64	1.63	0.58
60	20,020	8540	134.4	1.63	1.65	−0.98

8. Conclusions

In the present work, a preliminary aerothermochemical analysis of the axisymmetric flow field around Orion capsule was carried out at high altitude Mars entry. The analysis provided information about local quantities

(temperature, pressure and heat flux) in the flow field and on the capsule surface. The study also took into account the effects of surface catalyticity, and was carried out in the altitude interval 60–100 km by two codes for the computation of:

- (1) the capsule entry trajectory by means of integration of the equations of the capsule dynamics. Both direct and orbit entry trajectories were considered. The GRAM-2001 provided the atmospheric parameters.
- (2) the flow field around the capsule, local quantities along the TPS and drag. Computations were carried out by the DSMC code DS2V-4.5, 64 bits.

The analysis relied on four test conditions: 1) chemically inert gas, 2) chemically active gas and non catalytic surface, 3) chemically active gas and fully catalytic surface, 4) chemically active gas and insulated surface. Temperature at the stagnation point, where the most critical conditions occur, increases, from $h = 100$ km to $h = 60$ km, by one order of magnitude while pressure and heat flux increase by three orders of magnitude. The increase of heat flux due to surface recombination reactions was also evaluated. As expected, direct entry velocity, being higher than orbit entry velocity, involves higher values of all local quantities. However, due to the principle of independence of Mach number, the values of the corresponding coefficients, including the global drag coefficient, do not show meaningful variations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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